# Practical No 10

## A) Create a PWA with a web App Manifest file including:

- **App name**
- **Icons**
- **Start URL**
- **Display mode**
- **Implement Push Notifications in a PWA.**

**Project Name: PushNotification**
**Project Type: ASP.NET Core Empty Web App**
**Program Code:**

### I. Filename: index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>My PWA</title>
    <link rel="manifest" href="manifest.json">
    <meta name="theme-color" content="#0d6efd">
</head>
<body>

    <h1>My Progressive Web App</h1>

    <button onclick="showNotification()">
        Show Notification
    </button>

    <script src="app.js"></script>

</body>
</html>
```

### II. Filename: manifest.json

```
{
    "name": "My PWA Application",
    "short_name": "MyPWA",
    "start_url": "/index.html",
    "display": "standalone",
    "background_color": "#ffffff",
    "theme_color": "#0d6efd",
    "icons": [
        {
            "src": "icons/ppp1.png",
            "sizes": "256x256",
            "type": "image/png"
        },
        {
            "src": "icons/ppp2.png",
            "sizes": "512x512",
            "type": "image/png"
        }
    ]
}
```

## III. Filename: service-worker.js

```
self.addEventListener('install', event => {
    console.log('Service Worker Installed');
});

self.addEventListener('activate', event => {
    console.log('Service Worker Activated');
});

self.addEventListener('fetch', event => {
    event.respondWith(fetch(event.request));
});
```

## IV. Filename: app.js

```
if ('serviceWorker' in navigator) {
    navigator.serviceWorker.register('service-worker.js')
        .then(() => console.log('Service Worker Registered'));
}

function showNotification() {
    Notification.requestPermission().then(permission => {
        if (permission === 'granted') {
            navigator.serviceWorker.ready.then(registration => {
                registration.showNotification('PWA Notification', {
                    body: 'Push Notification Working Successfully!',
                    icon: 'icons/ppp1.png'
                });
            });
        }
    });
}
```

## V. Filename: Program.cs

```
var builder = WebApplication.CreateBuilder(args);

var app = builder.Build();

app.UseDefaultFiles();

app.UseStaticFiles();

app.Run();
```

## B) E-Commerce PWA with Product Listing and Offline Cart To develop an E-Commerce Progressive Web Application with product listing and offline cart functionality using Service Worker and local storage.

**Project Name: ECommercePWA**

**Project Type: ASP.NET Core Empty Web App**

**Program Code:**

## I. Filename: index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>E-Commerce PWA</title>

    <link rel="manifest" href="manifest.json">
    <meta name="theme-color" content="#0d6efd">

    <style>
        body {
            font-family: Arial;
            margin: 20px;
        }

        .container {
            display: flex;
            gap: 30px;
        }

        .cart-section {
            width: 30%;
            border: 1px solid gray;
            padding: 15px;
            min-height: 300px;
        }

        .product-section {
            width: 70%;
        }

        .products {
            display: flex;
            flex-wrap: wrap;
            gap: 15px;
        }

        .product {
            border: 1px solid gray;
            padding: 10px;
            width: 200px;
        }

        button {
            padding: 5px 10px;
            cursor: pointer;
        }
    </style>
</head>

<body>

    <h1>E-Commerce PWA</h1>

    <div class="container">

        <div class="cart-section">
            <h2>Cart Items</h2>
            <ul id="cartList"></ul>
        </div>

        <div class="product-section">
            <h2>Products</h2>

            <div class="products">

                <div class="product">
                    <h3>Laptop</h3>
                    <p>Price: ₹ 50000</p>
                    <button onclick="addToCart('Laptop')">
                        Add to Cart
```

```
                    </button>
                </div>

                <div class="product">
                    <h3>Mobile</h3>
                    <p>Price: ₹ 20000</p>
                    <button onclick="addToCart('Mobile')">
                        Add to Cart
                    </button>
                </div>

                <div class="product">
                    <h3>Headphones</h3>
                    <p>Price: ₹ 3000</p>
                    <button onclick="addToCart('Headphones')">
                        Add to Cart
                    </button>
                </div>

            </div>
        </div>

    </div>

    <script src="app.js"></script>

</body>
</html>
```

## II. Filename: manifest.json

```
{
    "name": "E-Commerce PWA",
    "short_name": "ShopPWA",
    "start_url": "/index.html",
    "display": "standalone",
    "background_color": "#ffffff",
    "theme_color": "#0d6efd",
    "icons": [
        {
            "src": "icons/ppp1.png",
            "sizes": "256x256",
            "type": "image/png"
        },
        {
            "src": "icons/ppp2.png",
            "sizes": "512x512",
            "type": "image/png"
        }
    ]
}
```

### III. Filename: service-worker.js

```
const CACHE_NAME = 'ecommerce-cache-v1';

const urlsToCache = [
    '/',
    '/index.html',
    '/app.js',
    '/manifest.json',
    '/icons/ppp1.png',
    '/icons/ppp2.png'
];

self.addEventListener('install', event => {
    event.waitUntil(
        caches.open(CACHE_NAME)
            .then(cache => {
                return cache.addAll(urlsToCache);
            })
    );
});

self.addEventListener('fetch', event => {
    event.respondWith(
        caches.match(event.request)
            .then(response => {
                return response || fetch(event.request);
            })
    );
});
```

### IV. Filename: app.js

```
if ('serviceWorker' in navigator) {
    navigator.serviceWorker.register('service-worker.js')
        .then(() => console.log('Service Worker Registered'));
}

let cart = JSON.parse(localStorage.getItem('cart')) || [];

displayCart();

function addToCart(product) {
    cart.push(product);
    localStorage.setItem('cart', JSON.stringify(cart));
    displayCart();
}

function displayCart() {
    let cartList = document.getElementById('cartList');

    cartList.innerHTML = "";

    cart.forEach(item => {
        let li = document.createElement('li');
        li.textContent = item;
        cartList.appendChild(li);
    });
}
```

### V. Filename: Program.cs

```
var builder = WebApplication.CreateBuilder(args);

var app = builder.Build();

app.UseDefaultFiles();

app.UseStaticFiles();

app.Run();
```